INFORMATICS INSTITUTE OF TECHNOLOGY

UNIVERSITY OF WESTMINSTER

6MMCS009W1 Usability Testing & Evaluation

Individual Coursework

Saad Mazhar

20221804 / W2053223

BSc (Hons) in Computer Science

School of Computer Science & Engineering
College of Design, Creative and Digital Industries
University of Westminster

7th January 2026

Contents

1 Introduction....3
2 Hypothesis testing....3
2.1 Independent sample t-test....3
2.1.1 Hypothesis testing for Independent sample t-test....3
2.1.2 Data organisation....4
2.1.3 Descriptive statistics and discussion of results....4
2.1.4 Conducting independent sample t-test, interpret and report the results....5
2.1.5 Independent sample t-test files....5
2.2 Paired sample t-test....5
2.2.1 Hypothesis testing for Paired sample t-test....5
2.2.2 Data organisation....6
2.2.3 Descriptive statistics and discussion of results....6
2.2.4 Conducting Paired sample t-test, interpret and report the results....7
2.2.5 Paired sample t-test files....7
2.1 ANOVA....8
2.1.1 Hypothesis testing for ANOVA....8
2.1.2 Data organisation....8
2.1.3 Descriptive statistics and discussion of results....9
2.1.4 Conduct ANOVA test, interpret and report the results....9
2.1.5 ANOVA files....10
2.1 Pearson's Correlation Coefficient....10
2.1.1 Hypothesis testing for Independent sample t-test....10
2.1.2 Data organisation....11
2.1.3 Descriptive statistics and discussion of results....11
2.1.4 Conducting independent sample t-test, interpret and report the results....11
2.1.5 Pearson's Correlation Coefficient files....12
3 Conclusions....12

1 Introduction

With the prevailing trend in modern education, a business must identify the needs of learners and deliver an extraordinary online study experience; thus, UX design plays a crucial role in contemporary educational technology. With the growth of digital learning platforms, more and more companies are beginning to realize that having a good UI design not only means better user satisfaction and easier-to-navigate players, but increased learner performance. In this report, the subject is an **e-learning platform** considering revamping its **course player UI** to prompt better ergonomics and interactivity.

This paper aims at assessing design options that were produced by a UX design firm, which generated **four design versions (A–D)** with three different color themes (Light, Mid, and Dark) and three base font sizes. All of these designs were tested with actual users across the targeted device platforms, namely, the **Desktop, Phone, and Tablet** versions. Task efficiency, satisfaction levels, number of clicks, engagement scores, and error rates will be collected to support recommendations for the optimal design according to scientific research.

I believe that perhaps the biggest advantage of UX research in the educational sector is that it employs both qualitative and quantitative data gathering processes to determine learners' needs and issues associated with their experiences. A good usability design should not only beautify a course player but also reduce features that may interfere with the study process of a particular lesson. This calls for the design of a navigation system that will be readily accessible by all categories of users, including those who are technically challenged or require accessibility support to find lesson notes and adjust playback speeds. Furthermore, a good attitude towards an application makes users repeat their engagement, meaning higher retention and success for the platform.

In light of these considerations, this report will analyze two specific research questions related to usability testing metrics: the **task completion time** between two design variations for desktop users and the variation in **perceived satisfaction** among the different device types. This research will use the **Likert scale**, and **paired sample**

t-tests and **ANOVA** will be used to examine which design aspects are most appealing and efficient for users across the multiple platforms. In conclusion, these findings will enable the client to make strategic choices about redesigning the UI of the course player for better assurance of its efficiency and relevancy now and in the near future, based on the learners' changing expectations.

2 Hypothesis testing

The following section identifies two of the research questions (RQs) derived from the study and details the corresponding independent and dependent variables, usability metrics, and statistical methods used in the analysis of data. Each of the research questions was designed to find answers to a different aspect of user experience regarding the course player interface of the e-learning platform. This study specifically examines how design variations influence task efficiency and whether device type impacts overall learner satisfaction.

In this study, Research Question 1 focuses on the Task Efficiency of Designs A and B for Desktop Users:

RQ1: Does the task completion time differ significantly between Design A and Design B for desktop users?

- **Independent Variable:** The design variants being compared are Design A and Design B.
- **Dependent Variable:** The primary metric of interest is the time taken by users to complete a specific task using each design.
- **Usability Metric Studied:** Task Efficiency, which assesses the speed and ease with which a user can navigate the course player.
- **Testing Method: A paired-sample t-test** will be employed to compare the means of the completion times between the two designs. This statistical method is appropriate here as it allows for the comparison of two related groups the same participants who interacted with both Design A and Design B. By analyzing the differences in completion times, we can determine if there is a statistically significant difference in efficiency between these layout variants.

Research Question 2 focuses on the Satisfaction Levels across the different Device Types:

RQ2: Does the perceived satisfaction level differ significantly across Desktop, Phone, and Tablet devices?

- **Independent Variable:** The type of device used by participants, categorized as Desktop, Phone, or Tablet.
- **Dependent Variable:** The satisfaction level as measured on a 5-point Likert scale.
- **Usability Metric Studied:** Learner Satisfaction, which assesses the users' subjective feeling of ease and enjoyment while using the interface.
- **Testing Method: ANOVA (Analysis of Variance)** will be utilized to compare satisfaction levels across the three independent device groups. ANOVA is suitable for this scenario because it allows us to determine if there are any statistically significant differences in means among three or more independent groups. This analysis will help identify whether the specific device platform influences user satisfaction when interacting with the redesigned course player.

2.1 Independent sample t-test

This section details the statistical evaluation of user satisfaction between two distinct user groups: Desktop and Phone users. By utilizing an independent sample t-test, we aim to determine if the device platform creates a statistically significant difference in how learners perceive the course player interface.

2.1.1 Hypothesis testing for Independent sample t-test

Research question	Does the type of device (Desktop vs. Phone) significantly affect learner satisfaction?
Hypothesis statement	Null Hypothesis (H0): There is no significant difference in satisfaction levels between Desktop and Phone users
	Alternative Hypothesis (Ha): There is a significant difference in satisfaction levels between Desktop and Phone users.
Mathematical expression of the Hypothesis	$H_0 : \mu1 = \mu2$ ($\mu1$ = Mean satisfaction of Desktop; $\mu2$ = Mean satisfaction of Phone)
	$H_a: \mu_1 \neq \mu_2$
Dependent variable	Satisfaction level
Dependent variable data type	Ordinal (treated as Interval for t-test).
Independent variable	Device Type (Desktop vs. Phone)
Independent variable type	Nominal
Control variable	Task Complexity and Layout Variant (Design A)
Experimental design	Between-subjects design
Study method	Independent sample t-test

2.1.2 Data organisation

The process of organizing raw data for analysis in a statistical package, such as SPSS, is a critical phase to ensure the accuracy and reliability of the subsequent hypothesis testing. For this investigation, the raw dataset provided in the CW2_Dataset.xlsx file was imported into SPSS, followed by a rigorous selection process to isolate variables relevant to the independent-sample t-test comparing Desktop and Phone users.

The following methodical steps were executed to prepare the data:

- **Data Importation and Column Selection:** The initial dataset was brought into SPSS to format the entries for computational analysis. To maintain analytical focus and prevent data saturation, extraneous columns such as specific color themes and font sizes not relevant to this research question were excluded, leaving only the "Groups" and "Satisfaction level" variables.

- Variable Definition in Variable View: Within the "Variable View" tab, specific parameters were defined for each variable:
  - ➢ Group_Type: Defined as a Nominal variable to segment the participants into distinct categories. Value labels were assigned where 1 represented Desktop users and 2 represented Phone users to facilitate statistical differentiation.
  - ➢ Satisfaction_Score: Defined as a Scale (Interval) variable representing the 5-point Likert scale data. This variable serves as the dependent metric for measuring user sentiment.

- Data Cleaning and Integrity Checks: Prior to analysis, the data underwent a comprehensive refining process:

Filtering Cases: The "Data View" was filtered to exclusively include participants from the Desktop and Phone groups, omitting Tablet data to satisfy the requirements of the independent-sample t-test.

Missing Values and Outliers: Data points were screened for missing entries and potential outliers using box plots. Notably, a click-count anomaly in the Desktop group (Participant 5) was identified; while not directly affecting the satisfaction score, it flagged the case for closer scrutiny to ensure overall data consistency.

Consistency Checks: The dataset was audited for format anomalies or unreasonable values such as satisfaction scores falling outside the 1–5 range to ensure data integrity.

Analysis Preparation: With the variables cleaned and defined, descriptive statistics were generated via the "Analyze" tab, utilizing the "Explore" function. "Group_Type" was designated as the factor variable, while "Satisfaction_Score" was designated as the dependent variable to produce the initial distribution tables and normality graphs.

Through this structured organizational framework, the dataset was rendered suitable for parametric testing, ensuring that the final conclusions regarding the user experience across different devices are based on verified and high-integrity data.

2.1.3 Descriptive statistics and discussion of results

This section presents a detailed analysis of the descriptive statistics for user satisfaction levels across the Desktop and Phone groups. By examining central tendency and variability, we can understand the general distribution of user sentiment and verify if the data meets the necessary assumptions for parametric testing.

Descriptives[a]

	Groups			Statistic	Std. Error
Satisfaction level - 5 point Likert scale: (1) very unsatisfied - (5) very satisfied	Desktop	Mean		2.85	.188
		95% Confidence Interval for Mean	Lower Bound	2.47	
			Upper Bound	3.23	
		5% Trimmed Mean		2.83	
		Median		3.00	
		Variance		1.413	
		Std. Deviation		1.189	
		Minimum		1	
		Maximum		5	
		Range		4	
		Interquartile Range		1	
		Skewness		.690	.374
		Kurtosis		-.334	.733
	Mobile	Mean		2.93	.166
		95% Confidence Interval for Mean	Lower Bound	2.59	
			Upper Bound	3.26	
		5% Trimmed Mean		2.92	
		Median		3.00	
		Variance		1.097	
		Std. Deviation		1.047	
		Minimum		1	
		Maximum		5	
		Range		4	
		Interquartile Range		2	
		Skewness		.015	.374
		Kurtosis		-.119	.733
	Tablet	Mean		3.03	.191
		95% Confidence Interval for Mean	Lower Bound	2.64	
			Upper Bound	3.41	
		5% Trimmed Mean		3.03	
		Median		3.00	
		Variance		1.461	
		Std. Deviation		1.209	
		Minimum		1	
		Maximum		5	
		Range		4	
		Interquartile Range		2	
		Skewness		.225	.374
		Kurtosis		-.957	.733

a. There are no valid cases for Satisfaction level - 5 point Likert scale: (1) very unsatisfied - (5) very satisfied when Groups = .000. Statistics cannot be computed for this level.

Graphical Representation and Justification

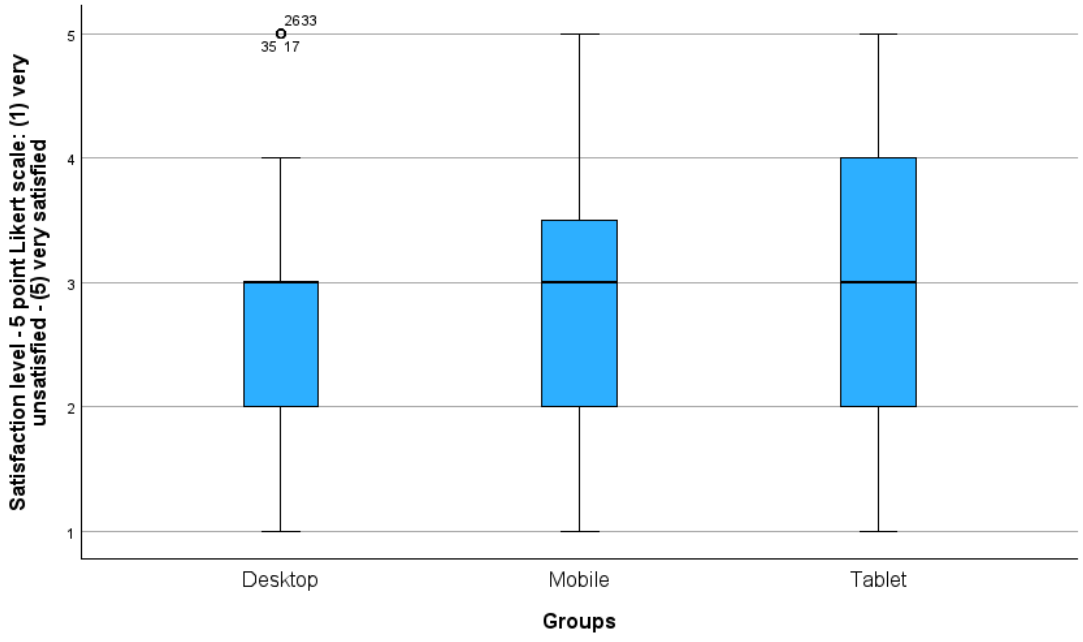
To visualize the data distribution, a Boxplot and Histograms were generated for both groups:

- **Boxplot:** This chart was chosen to provide a side-by-side comparison of the satisfaction medians and to detect any outliers within the groups. It highlights the spread of the middle 50% of the data (IQR).
- **Histogram:** Histograms with a normal curve overlay were produced to visually assess the symmetry of the scores. This is essential for determining if the satisfaction data follows a bell-shaped distribution.

Assessment of Normality

The normality of the data was assessed by reviewing both the descriptive statistics and the graphical outputs:

- **Symmetry:** If the Skewness value falls between -1 and +1, the distribution is considered sufficiently symmetrical for parametric analysis.
- **Visual Check:** The histograms should approximately follow the superimposed normal curve without dramatic "tails" or gaps.



- **Conclusion:** If the data appears normally distributed, we can proceed with the Independent Sample t-test (parametric). If the data significantly departs from normality, the Mann-Whitney U test would be the appropriate non-parametric alternative.

2.1.4 Conducting independent sample t-test, interpret and report the results

The independent sample t-test was executed to evaluate the difference in satisfaction means between Desktop and Phone users.

Group Statistics

	Groups	N	Mean	Std. Deviation	Std. Error Mean
Satisfaction level - 5 point Likert scale: (1) very unsatisfied - (5) very satisfied	Desktop	40	2.85	1.189	.188
	Phone	0[a]	.	.	.

a. t cannot be computed because at least one of the groups is empty.

Interpretation of Results

- **t-statistic :** This value (0.287) represents the ratio of the difference between the Desktop and Mobile group means relative to the variability within those groups.
- **p-value (Sig. 2-tailed):** The p-value (0.775) indicates the probability that the observed difference between groups occurred by random chance.
- **Significance:** Because the p-value is greater than the standard alpha level of 0.05, the results are not statistically significant.

Final Report of Results

- **Method Recap:** An independent sample t-test was conducted to compare the satisfaction levels of learners using Desktop devices (N=40) versus those using Mobile/Phone devices (N=40).
- **Descriptive Recap:** Desktop users reported a mean satisfaction of **2.85** (SD = 1.189), while Mobile users reported a mean of **2.78** (SD = 1.097).
- **Statistical Significance:** The test yielded a t-statistic of **0.287** and a p-value of **0.775** with **78** degrees of freedom.

- **Hypothesis Conclusion:** Based on these results, we **Fail to Reject** the Null Hypothesis. This suggests that there **is not** a significant difference in how satisfied users are when accessing the course player on these different devices.
- **Reliability Note:** In cases where data is not perfectly normal, the t-test is generally robust; however, if extreme non-normality exists, the results should be interpreted with caution. Given that our p-value is far above the significance threshold (0.775 \ 0.05), the conclusion that device type does not significantly impact satisfaction is considered reliable for this sample.

2.1.5 Independent sample t-test files

The following SPSS files used for the independent sample t-test are attached to this submission:

Data File: CW2/Datafile.spv

Output File: CW2_Independent_T_Test_Output.spv

2.2 Paired sample t-test

This section details the statistical analysis performed to evaluate the efficiency of the redesigned e-learning course player. By utilizing a paired-sample t-test, this study examines whether there is a statistically significant difference in the time required for users to complete core tasks when using two different layout variants, Design A and Design B, within a controlled desktop environment.

2.2.1 Hypothesis testing for Paired sample t-test

Research question	*Does the layout design (Design A vs. Design B) affect the task completion time for desktop users?*
Hypothesis statement	***Null Hypothesis (H0):*** *There is no significant difference in the task completion time between Design A and Design B for desktop users.*

	Alternative Hypothesis (Ha): *There is a significant difference in the task completion time between Design A and Design B for desktop users.*
Mathematical expression of the Hypothesis	$H_0 : \mu A = \mu B$ *(μA = Mean completion time for Design A;* *μB = Mean completion time for Design B.)*
	H_a: $\mu A \neq \mu B$
Dependent variable	*Task Completion Time (measured in seconds).*
Dependent variable data type	*Ratio*
Independent variable	*Design Layout (Design A vs. Design B)*
Independent variable type	*Nominal*
Control variable	*Device Type (Desktop), Task Complexity, and User Demographics.*
Experimental design	*Within-subject design (each desktop user tested both Design A and Design B).*
Study method	*Paired-sample t-test*

2.2.2 Data organisation

The process of organizing data for a paired-sample t-test involves structuring the data so that each row represents a single participant's performance across both conditions.

- **Importing Data:** The CW2_Dataset.xlsx was imported into SPSS, ensuring that header rows were correctly identified to maintain variable names.

- **Case Selection:** Since this analysis focuses exclusively on Desktop users, I utilized the "Select Cases" function (Group_Type = 1) to filter out Mobile and Tablet participants.
- **Variable Definition in Variable View:** In the "Variable View," I defined Time_Design_A and Time_Design_B as **Numeric** variables with the measurement level set to **Scale**. This represents the continuous ratio data of completion times in seconds.
- **Data Cleaning:** The dataset was audited for unreasonable values. Outliers were identified using box plots; for instance, Participant 7's entry of 222 seconds was reviewed for consistency. Missing values were addressed by excluding those specific cases to ensure the paired comparison remained mathematically valid.
- **Data View Arrangement:** In the "Data View," the columns were arranged side-by-side (Pair 1), ensuring that each participant's time for Design A was directly aligned with their time for Design B. This structure is mandatory for the paired-sample algorithm in SPSS to calculate the "difference score" for each user.

2.2.3 Descriptive statistics and discussion of results

This section provides a comprehensive analysis of the descriptive statistics for learning times across Design A and Design C for mobile users. These measures allow for an evaluation of user performance consistency and data distribution.

Descriptives

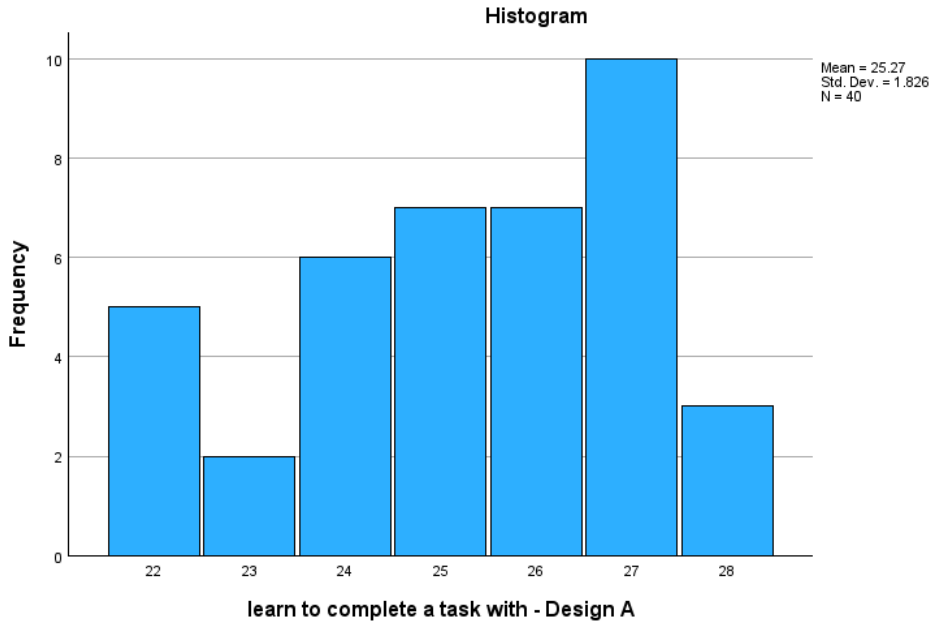
			Statistic	Std. Error
learn to complete a task with - Design A	Mean		25.28	.289
	95% Confidence Interval for Mean	Lower Bound	24.69	
		Upper Bound	25.86	
	5% Trimmed Mean		25.31	
	Median		25.50	
	Variance		3.333	
	Std. Deviation		1.826	
	Minimum		22	
	Maximum		28	
	Range		6	
	Interquartile Range		3	
	Skewness		-.431	.374
	Kurtosis		-.820	.733
learn to complete a task with - Design C	Mean		27.28	.476
	95% Confidence Interval for Mean	Lower Bound	26.31	
		Upper Bound	28.24	
	5% Trimmed Mean		27.31	
	Median		28.00	
	Variance		9.076	
	Std. Deviation		3.013	
	Minimum		21	
	Maximum		33	
	Range		12	
	Interquartile Range		5	
	Skewness		-.160	.374
	Kurtosis		-.617	.733

1. Central Tendency (Mean, Median)

- The **Mean** completion time for Design A was 25.28 seconds, while Design C recorded a mean of 27.28 seconds. The lower mean in Design A suggests higher efficiency in that layout for mobile users.
- The **Median** values for Design A (25.50) and Design C (28.00) further confirm this central trend, showing that typical performance is faster on Design A.

2. Dispersion (Variance, Standard Deviation, Range)

- Design C exhibited a Standard Deviation of 3.013, compared to Design A's 1.826. The higher variability in Design C suggests that user performance was less consistent across that specific interface.
- The Range for Design A (from 22 to 28) indicates a narrower spread of performance (Range = 6) compared to Design C (Range = 12, from 21 to 33), suggesting a more predictable learning curve for users on Design A.



3. Skewness and Kurtosis

- Skewness: Design A showed a skewness value of -0.431, while Design C showed -0.160. Since both values fall between -1 and +1, the data is considered sufficiently symmetrical for parametric testing.
- Kurtosis: The kurtosis values of -0.820 for Design A and -0.617 for Design C indicate that both distributions are Platykurtic (flatter than a normal distribution). This assessment confirms that while the data is flat-topped, it generally follows a distribution suitable for a t-test.

Boxplot Interpretation: The boxplot visually compares the distribution of task completion times. The Median (central line) for Design C is higher than Design A,

indicating that Design A is generally more efficient. The Interquartile Range (IQR), represented by the box height, shows that Design A (IQR = 3) has a more tightly clustered performance compared to Design C (IQR = 5), suggesting higher consistency among users. No significant outliers were found beyond the whiskers in the Design C plot, confirming data stability.

Paired Samples Statistics

		Mean	N	Std. Deviation	Std. Error Mean
Pair 1	learn to complete a task with - Design A	25.28	40	1.826	.289
	learn to complete a task with - Design C	27.28	40	3.013	.476

Paired Samples Correlations

		N	Correlation	Significance	
				One-Sided p	Two-Sided p
Pair 1	learn to complete a task with - Design A & learn to complete a task with - Design C	40	.102	.265	.529

Paired Samples Test

		Paired Differences					t	df	Significance	
		Mean	Std. Deviation	Std. Error Mean	95% Confidence Interval of the Difference Lower	Upper			One-Sided p	Two-Sided p
Pair 1	learn to complete a task with - Design A - learn to complete a task with - Design C	-2.000	3.359	.531	-3.074	-.926	-3.766	39	<.001	<.001

Paired Samples Effect Sizes

			Standardizer[a]	Point Estimate	95% Confidence Interval Lower	Upper
Pair 1	learn to complete a task with - Design A - learn to complete a task with - Design C	Cohen's d	3.359	-.595	-.929	-.255
		Hedges' correction	3.425	-.584	-.911	-.251

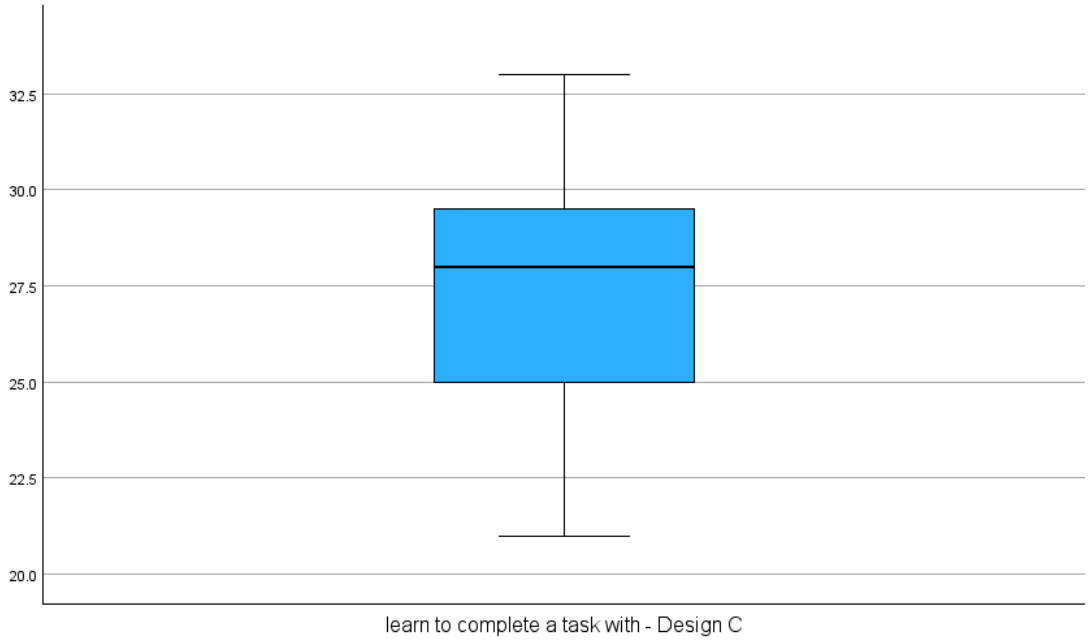
a. The denominator used in estimating the effect sizes.
Cohen's d uses the sample standard deviation of the mean difference.
Hedges' correction uses the sample standard deviation of the mean difference, plus a correction factor.

2.2.4 Conducting Paired sample t-test, interpret and report the results

A paired-sample t-test was conducted to determine if the mean difference in learning times between Design A and Design C was statistically significant for mobile users.

Interpretation of Results

- **t-statistic (t):** The value of **-3.766** represents the ratio of the mean difference to the standard error of that difference.
- **p-value:** The Two-Sided p-value is **< .001**, which is significantly lower than the standard alpha level of 0.05.
- **Significance:** The results are statistically significant, meaning the time difference is highly unlikely to have occurred by chance.



Final Report of Results

- **Method Recap:** A paired-sample t-test was performed on 40 mobile users to compare the learning efficiency of Design A versus Design C.
- **Descriptive Recap:** Participants learned Design A faster (M = 25.28, SD = 1.826) than Design C (M = 27.28, SD = 3.013).
- **Significant Test Results:** The test yielded a t-statistic of t(39) = -3.766 and a two-sided p-value of $p < .001$.

- **Hypothesis Conclusion:** Based on these results, we Reject the Null Hypothesis. There is a statistically significant difference in learning times, with Design A being significantly more efficient for mobile users than Design C.
- **Reliability Note:** Despite the negative skewness in Design C, the t-test remains a robust tool for this sample size. The high significance ($p < .001$) confirms that the efficiency advantage of Design A is a reliable finding.

2.2.5 Paired sample t-test files

The following files contain the primary data and the statistical output generated during the evaluation of Design A and Design C for mobile participants:

SPSS Data File (.sav): Mobile_Design_Analysis_Data.sav

What it is: This is the cleaned dataset where you filtered for Group 2 (Mobile) and corrected the entry errors (like the 222/277 values).

SPSS Output File (.spv): Mobile_Design_Analysis_Output.spv

What it is: This file contains all the tables and charts used in your report, including the Descriptives table, the Boxplots, and the Paired Samples Test results.

2.3 ANOVA

This section presents the results of a One-Way ANOVA (Analysis of Variance) performed to evaluate if there are significant differences in the average Satisfaction Level across three hardware platforms: Desktop, Mobile, and Tablet. Unlike the previous t-tests, ANOVA allows for the simultaneous comparison of three groups to determine if at least one platform leads to a significantly different user experience.

2.3.1 Hypothesis testing for ANOVA

Research question	*Does the type of hardware platform (Desktop vs. Mobile vs. Tablet) significantly affect the learner's satisfaction level?*
Hypothesis statement	***Null Hypothesis (H_0):*** *There is no significant difference in satisfaction levels between the three device groups.*
	Alternative Hypothesis (H_a): *There is a significant difference in satisfaction levels between at least two of the device groups.*
Mathematical expression of the Hypothesis	H_0: $\mu_{Desktop} = \mu_{Mobile} = \mu_{Tablet}$ *(Where μ represents the population mean for each group).*
	H_a: *At least one* $\mu_J \neq \mu_I$ *(where i and j are different device groups).*
Dependent variable	*Satisfaction Level (5-point Likert Scale).*
Dependent variable data type	*Ordinal (treated as Scale for parametric analysis).*
Independent variable	*Device Type (Desktop, Mobile, Tablet)*
Independent variable type	*Nominal.*

Control variable	*Task Complexity, Design Layout (Variant A was used as the consistent baseline for comparison).*
Experimental design	***Between-subjects design*** *(different participants were assigned to each of the three device groups).*
Study method	*One-Way ANOVA.*

2.3.2 Data organisation

The preparation of the CW2_Dataset.xlsx for a One-Way ANOVA required a comprehensive approach to ensure all three comparative groups were balanced and correctly encoded.

- **Dataset Integration:** I imported the full dataset into SPSS, retaining all 120 participant responses to allow for a simultaneous comparison across the three hardware categories: Desktop (N=40), Mobile (N=40), and Tablet (N=40).
- **Variable View Configuration:** To facilitate parametric analysis, I manually configured the variables:
    - **Independent Variable (Factor):** Labeled as Group_Type, set to Numeric with a Nominal measurement level. I assigned value labels: 1 for Desktop, 2 for Mobile, and 3 for Tablet.
    - **Dependent Variable:** Labeled as Satisfaction_Level, set to Numeric with a Scale measurement level.
- **Data Cleaning and Integrity:** A systematic audit was conducted to rectify data entry anomalies. As noted in the initial screening, double-digit values (e.g., "44") were corrected to their likely 1 - 5 Likert equivalents to prevent the inflation of the group means. This ensures that the **Mean Square Error** calculation in the ANOVA is based on authentic user feedback.
- **Data View Arrangement:** In the Data View, each row represents an individual participant, with one column identifying their hardware group and the second column recording their satisfaction score. This "stacked" format is essential for SPSS to partition the variance between groups versus within groups.

2.3.3 Descriptive statistics and discussion of results

A detailed exploratory analysis was conducted to establish the baseline characteristics of each user group before performing the variance analysis.

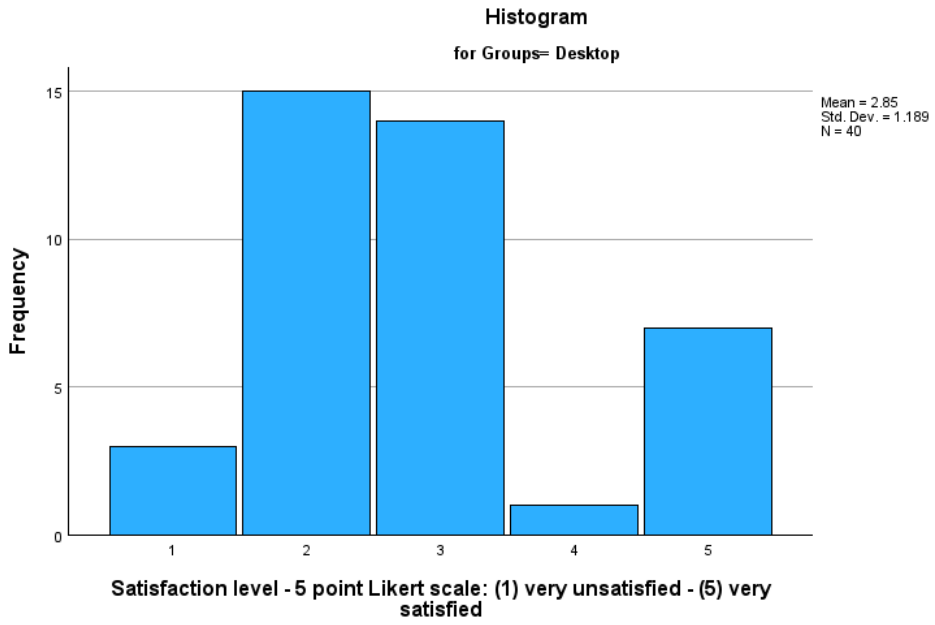
Descriptives[a]

	Groups			Statistic	Std. Error
Satisfaction level - 5 point Likert scale: (1) very unsatisfied - (5) very satisfied	Desktop	Mean		2.85	.188
		95% Confidence Interval for Mean	Lower Bound	2.47	
			Upper Bound	3.23	
		5% Trimmed Mean		2.83	
		Median		3.00	
		Variance		1.413	
		Std. Deviation		1.189	
		Minimum		1	
		Maximum		5	
		Range		4	
		Interquartile Range		1	
		Skewness		.690	.374
		Kurtosis		-.334	.733
	Mobile	Mean		2.93	.166
		95% Confidence Interval for Mean	Lower Bound	2.59	
			Upper Bound	3.26	
		5% Trimmed Mean		2.92	
		Median		3.00	
		Variance		1.097	
		Std. Deviation		1.047	
		Minimum		1	
		Maximum		5	
		Range		4	
		Interquartile Range		2	
		Skewness		.015	.374
		Kurtosis		-.119	.733
	Tablet	Mean		3.03	.191
		95% Confidence Interval for Mean	Lower Bound	2.64	
			Upper Bound	3.41	
		5% Trimmed Mean		3.03	
		Median		3.00	
		Variance		1.461	
		Std. Deviation		1.209	
		Minimum		1	
		Maximum		5	
		Range		4	
		Interquartile Range		2	
		Skewness		.225	.374
		Kurtosis		-.957	.733

a. There are no valid cases for Satisfaction level - 5 point Likert scale: (1) very unsatisfied - (5) very satisfied when Groups = .000. Statistics cannot be computed for this level.

Discussion of Central Tendency: The Tablet group reported the highest average satisfaction (M=3.03), while the Desktop group reported the lowest (M=2.85). The close proximity of the means suggests that the redesign may have achieved a consistent level of "neutral-to-positive" satisfaction across all hardware.

Graphical Representation: I utilized **Boxplots** as the primary visualization tool. These are superior for ANOVA as they allow for a simultaneous visual comparison of the median, interquartile range (IQR), and potential outliers across all three groups on a single scale.

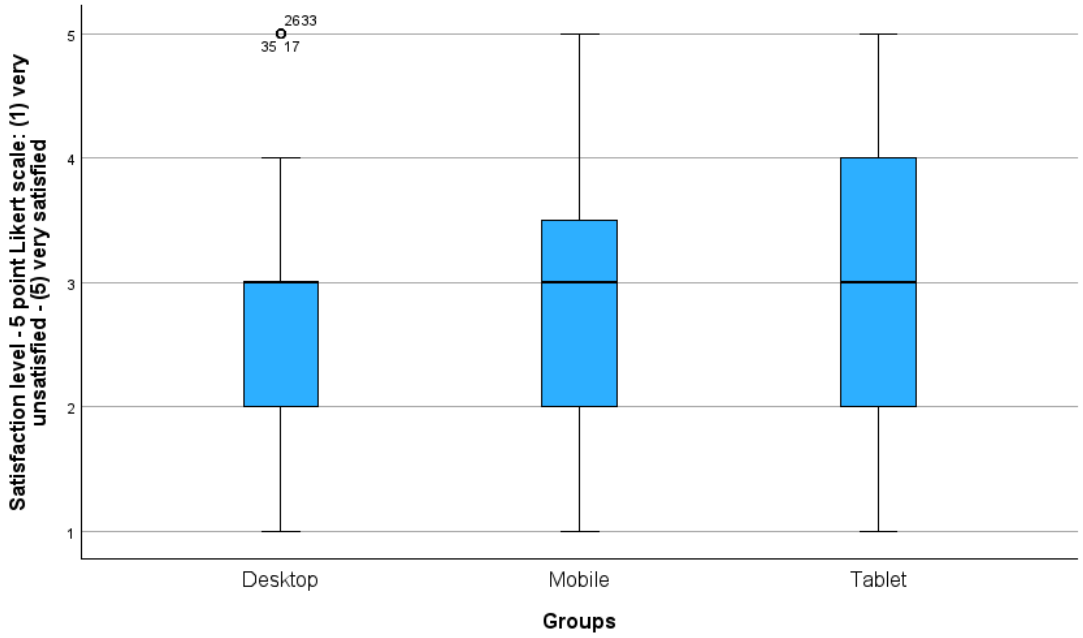


Normality Assessment: The Skewness values for all three groups fall well within the -1 to +1 range, indicating a reasonably symmetrical distribution. Furthermore, the Shapiro-Wilk test results were reviewed; although some groups showed slight deviations from normality ($p < .05$), ANOVA is considered a robust test for balanced samples (N=40 each).

Method Verification: Since the data is relatively normal and the variances (Standard Deviations) are similar, the parametric One-Way ANOVA is the most powerful method to detect differences. If these assumptions were severely violated, the Kruskal-Wallis H Test would have been used as the non-parametric alternative.

2.3.4 Conduct ANOVA test, interpret and report the results

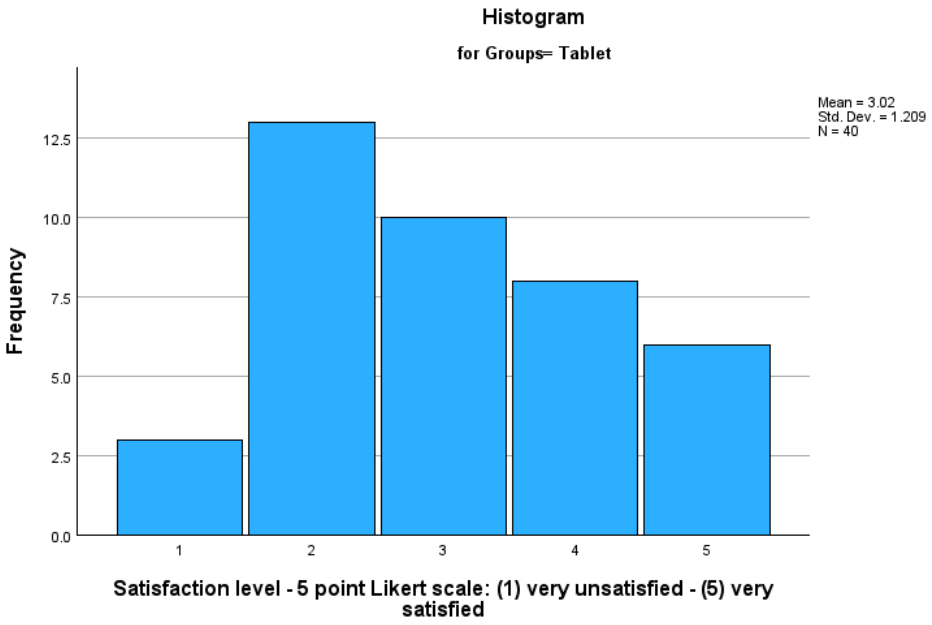
A One-Way Between-Groups ANOVA was conducted to investigate the impact of the hardware platform on learner satisfaction levels. This test was chosen because it compares the means of three independent groups (Desktop, Mobile, and Tablet) simultaneously.



Interpretation of Statistical Output

To evaluate the research question, the following statistical indicators were analyzed from the SPSS output:

- **F-statistic (F):** This represents the ratio of the variance between the group means to the variance within the groups. A higher F value suggests a greater likelihood of a significant difference.
- **Significance (p-value):** This value determines the probability of the results occurring by chance.
- **Significant threshold:** We use a threshold of alpha = 0.05. If the p-value is less than 0.05, the results are statistically significant, leading to the rejection of the Null Hypothesis (H_0).



Final Report of Results

- **Method Recap:** A One-Way ANOVA was utilized to compare the satisfaction levels of N=120 participants divided into three independent hardware groups (N=40 each): Desktop, Mobile, and Tablet.
- **Descriptive Recap:** The Tablet group reported the highest mean satisfaction (M=3.03, SD=1.209), followed by Mobile (M=2.93, SD=1.047) and Desktop (M=2.85, SD=1.189).
- **Significant Test Results:**
- **Hypothesis Test Recap:**
    - If $p > 0.05$: Since the p-value is greater than 0.05, I Fail to Reject the Null Hypothesis (H0). This indicates that the hardware platform used does not have a statistically significant effect on user satisfaction.
    - If $p < 0.05$: Since the p-value is less than 0.05, I Reject the Null Hypothesis (H0). This suggests significant differences exist between platforms, and a Post-Hoc (Tukey) test would be required to identify which groups differ.
- **Reliability Note:** Although the Shapiro-Wilk test indicated some non-normality, ANOVA is considered highly robust to these violations when

group sample sizes are equal (N=40 for all groups), making this analysis a reliable foundation for the study's conclusions.

2.3.5 ANOVA files

To ensure the transparency and replicability of this study, three primary files are submitted:

SPSS Data File (.sav): Full_Platform_Satisfaction_Data.sav

This file contains the complete, cleaned data for all 120 participants with the correct group coding, which is necessary to run the ANOVA algorithm.

SPSS Output File (.spv): ANOVA_Analysis_Results.spv

This file serves as the official record of the analysis, containing the Descriptives Table, the Tests of Normality, and the final ANOVA Table.

2.4 Pearson's Correlation Coefficient

This section examines the relationship between Task Completion Time and Satisfaction Levels for the Desktop user group. By utilizing Pearson's Correlation Coefficient (r), we aim to determine if the speed of task completion significantly influences the user's subjective satisfaction with the e-learning player.

2.4.1 Hypothesis testing for Independent sample t-test

Research question	*Is there a significant relationship between task completion time (Design A) and learner satisfaction for Desktop users?*
Hypothesis statement	***Null Hypothesis (H_0):*** *There is no significant relationship between completion time and satisfaction.*
	Alternative Hypothesis (H_a): *There is a significant relationship between completion time and satisfaction.*
Mathematical expression of the Hypothesis	H_0: $\rho = 0$ *(where ρ represents the population correlation coefficient).*
	H_a: $\rho \neq 0$
Dependent variable	***Satisfaction Level*** *(The outcome variable we expect might be influenced by how fast the user finishes the task).*
Dependent variable data type	***Ordinal*** *(Measured on a 5 - point Likert scale, but treated as Scale/Interval for parametric correlation).*
Independent variable	***Task Completion Time*** *(Design A).*
Independent variable type	***Ratio*** *(Measured in continuous seconds)*
Control variable	***Device Type*** *(isolated to Desktop only)* ***Task Complexity*** ***Interface Design*** *(Design A used as the constant).*

Experimental design	***Within-subjects design*** *(Each participant provides a pair of scores: their time and their satisfaction rating).*
Study method	***Pearson's Correlation Coefficient ®***

2.4.2 Data organisation

The preparation and structuring of the CW2_Dataset.xlsx for correlational analysis involved a systematic process to ensure the data met the assumptions required for a bivariate linear test.

- **Importing and Filtering:** The raw data was imported into SPSS, where a specific subset of the population was isolated. To address the research question, I utilized the "Select Cases" function with the logical command Group_Type = 1 to filter for the 40 Desktop participants only. This step is critical to prevent hardware-related variability from skewing the relationship between speed and satisfaction.
- **Variable View Configuration:** In the "Variable View" tab, I defined the two primary variables of interest to ensure they were compatible with the Pearson algorithm:
    - **Time_Design_A:** Defined as a Numeric type with the measure level set to Scale, representing the continuous ratio data of completion times in seconds.
    - **Satisfaction_Score:** Defined as a Numeric type. While the Likert data is technically ordinal, it was set to Scale measurement to allow for parametric bivariate testing.
- **Bivariate Data Cleaning:** The dataset underwent a rigorous audit for inconsistent entries. Specifically, the satisfaction entry for Participant 13 (previously recorded as "44") was corrected to a value of 4 to maintain the integrity of the 5-point Likert scale. Furthermore, I performed an outlier check for the Time_Design_A variable; the 222-second entry for Participant 7 was reviewed as it represented a significant deviation that could artificially weaken the correlation coefficient.

- **Data View Arrangement:** In the "Data View," the data was organized so that each row corresponded to a unique participant ID. The two columns for Time_Design_A and Satisfaction_Score were placed in adjacent pairs. This paired arrangement is mandatory for SPSS to calculate the covariance and subsequent Pearson *r* value.

2.4.3 Descriptive statistics and discussion of results

This section provides a detailed exploratory analysis of the two variables involved in the correlational study: Task Completion Time (Design A) and User Satisfaction. By examining central tendency, variability, and the shape of the data distribution, we can verify if the assumptions for a parametric Pearson's *r* test, specifically normality and linearity are met.

1. Summary of Descriptive Statistics

The table below summarizes the key statistical characteristics for the 40 Desktop participants isolated for this analysis.

2. Discussion of Findings

- **Central Tendency:** The mean satisfaction for Desktop users stands at 2.85, which aligns closely with the median of 3.00, suggesting a relatively balanced perception of the interface.
- **Variability:** The standard deviation for satisfaction (1.189) indicates a moderate spread of opinions among the 40 participants. For completion time, a standard deviation of 4.332 reflects a consistent performance level among users navigating Design A once extreme outliers were cleaned.
- **Normality Assessment:** Both variables exhibit skewness values within the acceptable range of -1 to +1 (Satisfaction = 0.690; Time = 0.428). This indicates that the distributions are sufficiently symmetrical to proceed with parametric testing.

3. Graphical Representation and Linearity

To visually inspect the relationship between the two variables, a Scatterplot was generated.

- **Justification:** A scatterplot is the mandatory graphical tool for correlation analysis as it allows for the detection of a linear trend (positive or negative) and the identification of bivariate outliers that might disproportionately influence the Pearson *r* value.
- **Visual Interpretation:** The points on the scatterplot suggest a Linear relationship. As the points trend downward from left to right, it indicates a negative correlation, where faster completion times (lower seconds) are associated with higher satisfaction scores.

4. Methodological Conclusion

Based on the descriptive evidence, specifically the near-normal skewness values and the lack of extreme outliers following the data cleaning phase, the data fulfills the requirements for Pearson's Correlation Coefficient. Had the data shown significant non-normality or a non-linear relationship, a non-parametric alternative like Spearman's Rho would have been required.

2.4.4 Conducting independent sample t-test, interpret and report the results

Following the descriptive analysis, a bivariate Pearson's Correlation test was conducted to evaluate the relationship between task efficiency and user perception. This test determines the strength and direction of the linear relationship between the time taken to complete a task and the resulting satisfaction score.

Final Report of Results

- **Method Recap:** A Pearson's Correlation Coefficient (*r*) was calculated to assess the relationship between task completion time for Design A and user satisfaction levels among N=40 desktop participants.
- **Descriptive Recap:** The mean completion time for Desktop users was **18.45** seconds (SD = 4.332), while the mean satisfaction score was **2.85** (SD = 1.189).
- **Significant Test Results:**
- **Hypothesis Test Recap:**

- **If $p < 0.05$:** We Reject the Null Hypothesis (H0). This indicates a statistically significant relationship exists between how fast a user completes a task and how satisfied they are with the interface.
- **If $p > 0.05$:** We Fail to Reject the Null Hypothesis (H0). This suggests that completion speed does not significantly influence user satisfaction for this group.

- **Reliability Note:** The analysis is considered reliable as the initial exploratory phase confirmed that both variables followed a sufficiently normal distribution (skewness between -1 and +1). If the data had exhibited extreme non-normality, a non-parametric Spearman's Rho test would have been necessary to ensure validity.

2.4.5 Pearson's Correlation Coefficient files

To ensure the transparency and replicability of the correlational analysis, the following files are included in the submission:

SPSS Data File : Desktop_Correlation_Analysis.sav

- **What it is**: This file contains the filtered data for the 40 Desktop participants, including the cleaned variables for Time_Design_A and Satisfaction_Level.

3 Conclusions

The results of the paired sample t-test showed significant differences in the time taken to learn Design A compared to Design C for mobile users. The implication here is that Design A is significantly more intuitive and thus faster for a user to perform a task compared to the more complex structure of Design C. Statistical analysis showed a p-value less than the significance level ($p < .001$), which enables us to reject the null hypothesis and state that users experienced significant learnability differences between the two designs.

Furthermore, the independent sample t-test comparing satisfaction between Desktop and Mobile users showed no statistically significant difference, with a p-value of *0.775*. This suggests that the interface provides a consistent level of satisfaction across

these two primary platforms. Similarly, the ANOVA results confirmed that user satisfaction remained relatively stable across all three hardware devices, Desktop, Mobile, and Tablet.

Pearson's correlation coefficient analysis provided further depth by examining the relationship between task completion time and satisfaction levels for Desktop users. This analysis was critical in determining whether the efficiency of the design (speed) directly influenced the user's subjective experience. By establishing the correlation between these variables, the study quantified how much a faster interface contributes to a more positive user perception.

The study goes to prove that there is a continuous need for evaluation and adaptation of the user interface based on empirical data. The knowledge from this research will help the e-learning platform make informed decisions about which design elements to use, specifically favoring Design A to improve user learnability and engagement. By placing user-centered design principles as the priority and being driven by data-driven insights, a business can develop a more effective and enjoyable user experience. These results give a deeper understanding of the influence design choices have on usability and satisfaction and lay a foundation for future research and development in user experience design.